

APPROVED METHOD:
IDENTIFYING OPERATING FACTORS
FOR INSTALLED HIGH INTENSITY
DISCHARGE LUMINAIRES
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-61-20(R2023)

**APPROVED METHOD:
IDENTIFYING OPERATING FACTORS
FOR INSTALLED HIGH INTENSITY
DISCHARGE LUMINAIRES
AN AMERICAN NATIONAL STANDARD**

Publication of this Committee Report
has been approved by the IES.
Suggestions for revisions
should be directed to the IES.

**Prepared by
The IES Testing Procedures Committee**



Copyright 2023 by the Illuminating Engineering Society.

Approved by the IES Standards Committee, December 19, 2022, as a Transaction of the Illuminating Engineering Society.

Approved April 28, 2023, as an American National Standard.

All rights reserved. No part of this publication may be reproduced in any form, in any electronic retrieval system or otherwise, without prior written permission of the IES.

Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

IES Standards are developed through committee consensus and produced by the IES Office in New York. Careful attention is given to style and accuracy. If any errors are noted in this document, they should be forwarded to the Director of Standards, at standards@ies.org or the above address for verification and correction. The IES welcomes and urges feedback and comments.

Printed in the United States of America.

ISBN# 978-0-87995-212-9

DISCLAIMER

IES publications are developed through the consensus standards development process approved by the American National Standards Institute. This process brings together volunteers representing varied viewpoints and interests to achieve consensus on lighting recommendations. While the IES administers the process and establishes policies and procedures to promote fairness in the development of consensus, it makes no guaranty or warranty as to the accuracy or completeness of any information published herein.

The IES disclaims liability for any injury to persons or property or other damages of any nature whatsoever, whether special, indirect, consequential or compensatory, directly or indirectly resulting from the publication, use of, or reliance on this document.

In issuing and making this document available, the IES is not undertaking to render professional or other services for or on behalf of any person or entity. Nor is the IES undertaking to perform any duty owed by any person or entity to someone else. Anyone using this document should rely on his or her own independent judgment or, as appropriate, seek the advice of a competent professional in determining the exercise of reasonable care in any given circumstances.

The IES has no power, nor does it undertake, to police or enforce compliance with the contents of this document. Nor does the IES list, certify, test or inspect products, designs, or installations for compliance with this document. Any certification or statement of compliance with the requirements of this document shall not be attributable to the IES and is solely the responsibility of the certifier or maker of the statement.

AMERICAN NATIONAL STANDARD

Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria have been met by the standards developer.

Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

The use of American National Standards is completely voluntary; their existence does not in any respect preclude anyone, whether that person has approved the standards or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standards.

The American National Standards Institute does not develop standards and will in no circumstances give an interpretation to any American National Standard. Moreover, no person shall have the right or authority to issue an interpretation of an American National Standard in the name of the American National Standards Institute. Requests for interpretations should be addressed to the secretariat or sponsor whose name appears on the title page of this standard.

CAUTION NOTICE: This American National Standard may be revised at any time. The procedures of the American National Standards Institute require that action be taken to reaffirm, revise, or withdraw this standard no later than five years from the date of approval. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute.

Prepared by the IES Testing Procedures Committee

KC Fletcher, *Chair*

Andrew Jackson, *Vice Chair*

David N. Randolph, *Secretary*

Members

C. K. Andersen	M. L. Grather	J. E. Leland	S. Mitsuhashi
R. S. Bergman	J. N. Hulett	K. M. Liepmann	E. Radkov
B. Boudreaux	P.-C. Hung	J. Lockner	M. B. Sapcoe
E. Bretschneider	Jianzhong Jiao	S. Longo	J. S. Swiernik
F. Carpenter	M. Kotrebai	L. Loudin	J. C. Vollers
D. Ellis	B. Kuebler	J. P. Marella	

Advisory Members

L. M. Ayers	P. Elizondo	G. McKee	M. Ruffin
R. P. Bergin	A. A. Feldman	C. C. Miller	M. Schneider
C. A. Bloomfield	J. Grandusky	Y. Ohno	T. Schneider
P.-T. Chou	K. J. Hemmi	E. Page	A. W. Serres
T. E. Cowan	R. E. Hlgley	D. Park	P. Sharma
P. Cruz	G. John	E. S. Perkins	R. W. Siddon
L. Davis	J. Juhasz	M. Piscitelli	D. Spicer
F. De Caso	R. Kelley	D. P. Ramer	G. A. Steinberg
J. J. Demirjian	J. D. Kramer	T. W. Rasinsky	M. Stevenson
M. E. Duffy	K. C. Lerbs	D. Rogers	L. Swainston
V. Eberhard	P. McCarthy	M. Royer	A. Thorseth

CONTENTS

1.0	Introduction and Scope	1
1.1	Introduction	1
1.2	Scope	1
2.0	Variables	1
2.1	Power Supply and Ballast Variations	1
2.1.1	Power Supply Voltage	1
2.1.2	Distribution System Impedance	2
2.1.3	Wave Shape of Voltage and Current	2
2.1.4	Ballasts	2
2.2	Lamp Variations	4
2.2.1	Light Output of the Lamp	4
2.2.2	Arc Tube Alignment	5
2.2.3	Arc Tube Temperature	6
2.2.4	Position Factor	6
2.2.5	Special Purpose Lamps	7
2.3	Luminaire Optical Variations	7
2.3.1	Reflector Materials	7
2.3.2	Reflector Shape	8
2.3.3	Refractors and Enclosures	8
2.3.4	Luminaire Assembly Tolerances	8
2.3.5	Luminaire Configurations and Options	8
2.4	Field Conditions	8
2.4.1	Mounting Height	8
2.4.2	Spacing	9
2.4.3	Aiming and Tilt	9
2.4.4	Topography and Obstructions	9
2.4.5	Surface Properties	9
2.5	Field Measurements	9
2.5.1	Meter Accuracy	9
2.5.2	Contaminant Conditions	10
2.5.3	Light Meter Positioning	10
2.5.4	Measurement Grid and Registration	10
2.5.5	Humidity and Temperature	11
2.5.6	Atmospheric Transmissivity	11
2.5.7	Weather and Site Conditions	11

3.0 General Considerations 11

4.0 Additional Reading 12

Annex A – Measurement and Recording Recommendations 12

Annex B – Measured vs. Predicted Performance Checklist 13

References 14

1.0 Introduction and Scope

1.1 Introduction

While lighting manufacturers and independent testing laboratories generally follow IES published procedures for testing HID luminaires under controlled laboratory conditions, measurements made in the field to confirm predicted performance can be influenced by a variety of factors. Under laboratory conditions, care is exercised in the control of the precision and accuracy of the photometric measurement conditions, including luminaire orientation, test distances, ambient temperature, electrical supply and regulation, and selection of lamps for optimal stability and arc tube geometry.

In the field, this level of control and accuracy in operating and measurement conditions is seldom possible, and measurements made of installed photometric performance can deviate from predicted performance levels. This Lighting Measurement (LM) document identifies many of the potential factors that apply to outdoor HID luminaires, including mercury vapor (Hg), metal halide (MH), and high pressure sodium (HPS), as well as low pressure sodium (LPS) lighting equipment utilizing core and coil ballasts for contemporary lamp wattage designations. The issues and factors associated with electronic ballasts for HID lamps have not been addressed in this revision of LM-61.

1.2 Scope

This LM outlines factors that can cause differences between calculated and measured illuminance and luminance values of outdoor high intensity discharge (HID) and low pressure sodium (LPS) lamps and luminaires. It does not offer solutions, nor does it quantify all of the possible variables that might be encountered. The relevant ANSI/IES Recommended Practice document should be consulted for specific design recommendations.

2.0 Variables

This document lists variables that can affect the results of an HID lighting design and outlines issues encountered

when comparing field measurements with predicted calculations using photometric data. It is possible that the tolerances for all the components could combine to produce a substantial difference in illuminance or luminance performance between laboratory and field conditions. The variables can be divided into five general categories:

- Power supply and ballast variations
- Lamp variations
- Luminaire optical variations
- Field conditions
- Field measurement

Each category is reviewed and its significance with respect to HID luminaire operation is explained.

2.1 Power Supply and Ballast Variations

SAFETY PRECAUTION: Only persons experienced in using measurement equipment under high voltage conditions should make measurements of electrical characteristics at the lamp. High primary and secondary voltages, high-voltage starting pulses for both HPS and MH lamps (up to 5,000 V), and high-voltage instant re-strike circuits (over 40 kV) present a serious shock hazard. It is critical that proper precautions also be taken to protect instruments from the high voltages and high-voltage starting pulses.

2.1.1 Power Supply Voltage. In the photometric laboratory, it is standard practice to maintain lamp watts at a constant value when HID lamps, or luminaires using them, are being photometered. This is typically accomplished by monitoring lamp power via a wattmeter and adjusting the supply voltage to the ballast until the lamp is stable at rated lamp watts. This establishes the luminaire performance and efficiency for the photometric report with a standardized procedure.

The predicted light output in the field assumes that a “nominal” electrical supply is used. In field installations, lamps may or may not operate at their rated wattage even though the ballasts are operated within the design limits of the electrical distribution system. The supply voltage at a luminaire is dependent on several electrical power system characteristics. Factors such as local time-variable power loads, transformer tap choices, and the length and gauge of the wiring runs can all

influence the supply voltage at a luminaire. Even under ideal conditions, power system voltage regulation is limited to 5%. Ballasts have a limited capability to regulate power to the lamp, and actual lamp wattage is influenced by the fixture's supply voltage where installed. There are several basic ballast types, each providing differing regulation capabilities and differing sensitivities to supply voltage variations.

Depending on lamp type and ballasting, a 1% change in supply voltage can cause a change in light output from 0.25% to 4.0%, as summarized in **Table 2-1**.

Table 2-1. Approximate Lumen Output Variation Due to 1% Supply Voltage Variation for Common 400-Watt Lamps

Ballast Type	Lamp Type	
	MH, PMH	HPS
Reactor	4.5%	3%
CWA	2.7%	1.8%
Reg Lag	1.4%	0.9%

For these reasons, it is important to know the power system conditions at the time of the photometric field measurements. Current, voltage, and wattage measurements can be made at ballasts and lamp sockets to determine whether the lamps are operating at rated conditions. The use of a recording voltmeter is recommended during the photometric measurements. If a recording voltmeter is not available, then periodic checks of the luminaire supply voltage should be made during the field measurements to be certain that any variations are recorded.

2.1.2 Distribution System Impedance. High distribution system impedance can cause conditions of low voltage, poor regulation, and excessive harmonics in a lighting system's power supply. A lighting circuit supplied by undersized distribution transformers can exhibit excessive source impedance. With increased use of pulse-start lamps and ballasts, consideration should be given to circuit sizing for the current demands created during hot-restrike conditions. Transformers with lower KVA ratings have increased series reactance for a given voltage rating. Distribution transformers have ambient air temperature ratings that affect the life of the transformer and the winding resistance. Maintaining a margin of safety on transformer temperature ratings

will help reduce system impedance. Excessive source harmonics or resonant conditions can affect source impedance because the system impedance is optimized for the fundamental operating frequency.

2.1.3 Wave Shape of Voltage and Current. Surges, transients, and non-sinusoidal wave forms with excessive harmonic content can be detrimental to lamps and ballasts. Typically, these effects result in decreased system life and/or increased lamp and ballast mortality. Grounded distribution systems of 120, 277, and 347 volts are usually preferred and result in smaller variations of wave shape to HID ballasts. When this type of supply is combined with non-isolated ballasts, such as auto-transformer and reactor types, satisfactory results are usually achieved with lower transients and harmonics in the power supply. On distribution systems where both supply lines are above ground potential, such as 208, 240, 480, and 600 volts, isolated constant-wattage ballasts operate with fewer problems. However, multiple faults to ground might cause fluctuations of power input, ballast burnout, variable lamp output and possible failures.

2.1.4 Ballasts. Ballast differences can be a significant factor creating inconsistencies between measurement and prediction. The ballast factor is the ratio of the lumen output produced when a given lamp is energized by a production ballast as compared to the light output produced when energized by a laboratory reference ballast.¹ ANSI standards for lamps and ballasts only ensure operation compatibility.² The ANSI standard ensures that the lamp will light, that the lamp will not be damaged electrically, and that the operating wattage is within a known tolerance. However, interaction between the lamp and ballast electrical properties is such that a specific combination determines the unit's operating wattage.

Many HID ballasts are not supplied as a matched set of components. If the coil and core and the capacitor are supplied independently rather than as a matched unit, the individual component tolerances can add to uncertainty in the operating wattage. Finally, there are many types of circuits commonly used for HID ballasts that vary in their ability to compensate for lamp and capacitor variations.

For reasons such as these, it is difficult to quantify the range of differences between rated lamp lumens and the initial operating lumens on a production ballast. ANSI specifications permit variation in arc voltage and on lamp lumens because of the corresponding change in the lamp's operating wattage.³ The range of lumen output variation expected due to ANSI-permitted ballast wattage variation at nominal values for lamp voltage, supply voltage, and capacitance is at $\pm 7.5\%$ for Hg lamps, $\pm 5\%$ for MH lamps, and $\pm 7.5\%$ for HPS lamps. The factors affecting these variations are described in the subsections that follow.

2.1.4.1 Design Variations. There are many general ballast types available, most of which are intended to deliver the correct input power to the lamp. (For further information on ballast types, refer to the American National Standards, C82 Series, Lamp Ballasts and Transformers.)

2.1.4.2 Core and Coil. Variation in the metallurgical composition of the steel causes variations in core losses and permeability, which affect lamp wattage. The probability of the number of turns in the coils varying is small, but it does happen, as does variation in the number of laminations in the core. There is also the potential for variation in shunt and air-gap geometry.

2.1.4.3 Capacitor. Capacitors may be found in the primary or secondary side of a ballast, as an aid to starting and as an active circuit component for power factor improvement. In many ballast circuits, capacitors also influence the lamp wattage. Replacement capacitors that match exactly in value and tolerance should be provided. The temperature of the capacitor may cause some changes in light output when comparing the photometric test unit with standard production items.

Published capacitor tolerances are normally $\pm 3\%$ of the specified value, although typical production variation may run near $\pm 1.5\%$ of specified values, with corresponding changes in operating wattage. Wattage variation due to the expected $\pm 3\%$ capacitance variation is typically 1.8 times the capacitor tolerance, for an expected wattage range of $\pm 5.4\%$ and a nearly proportional variation in lumen output.

2.1.4.4 Starting Aids. With normal starting aids for HPS lamps and some metal halide lamps, there is little probability that variations therein might affect lamp wattage unless the circuit is defective.⁴ Such a defect would in most cases render the starting circuitry inoperative. Some varieties of hot-restrike circuitry rely on an additional step-up transformer in series with the lamp circuit to reignite the lamp. The transformer might add impedance to the lamp circuit, resulting in reduced current and power to the lamp, and thereby reducing light output.

2.1.4.5 Wave Shape of Voltage and/or Current. The ballast regulation should maintain lamp wattage within the range specified by the lamp manufacturer, which should be the range listed by ANSI lamp specifications.³ The waveform of the lamp voltage and current should be in compliance with the lamp manufacturer's data, based on ANSI requirements where applicable, to ensure lamp operating stability and performance. Variations of light output and lamp life will result if these suggested limits are not followed. The crest factor (peak to RMS ratio) should also be within ANSI specifications. The ballast should limit the maximum lamp crest factor to the value listed in ANSI standards. Open circuit voltage and starting current during startup are also critical items to ensure timely lamp warmup without excessive electrode deterioration, which could result in accelerated lumen depreciation. Accelerated lumen depreciation could make field measurements inconsistent with predictions of maintained lumens as a function of operating hours.

2.1.4.5 Ballast Component Temperatures. Ballasts and capacitors have individual and combined thermal effects that influence the operating wattage of a luminaire. Their operating temperatures are a result of power dissipated in the luminaire, balanced by the convective, radiant, and conductive heat transfer of the components to the luminaire housing. Variations in ballast component assembly such as position, fastener tightness, contact surface flatness, and thermal compounds in the luminaire enclosures can result in luminaire-to-luminaire variations in light output.

Luminaires are typically photometered using relative photometry, with external reference reactors at rated wattage; therefore, the system-level electrical

temperature dependencies are *not* necessarily included as a part of the photometric testing's efficiency values. In the field, component temperatures can influence lamp operating wattages and field-measured light output. In a related manner, replacement components, such as capacitors and ballasts of different physical size or shape but of the same electrical rating, or those with only approximately the same ratings as the original combination, could also produce a light-output change due to changes in heat transfer properties. Crosschecking these issues may be necessary in instances where lamp operating wattage is found to be appreciably different than planned in the predictive analysis.

2.2 Lamp Variations

2.2.1 Light Output of the Lamp. Many luminaires are designed to operate with a variety of lamp options. Photometry is typically measured and published for each lamp option or a related group of lamp options. It is important to understand the impacts of the installed lamp selection on measured versus predicted performance. Within a given wattage and lamp type, such as 400-watt metal halide, the lamp manufacturers will produce a variety of lamp configurations, such as: clear and coated jackets, standard and reduced-size jackets, protected and unprotected arc tubes, universal and orientation-specific constructions (e.g., VBU or HOR), standard output and high output versions. Additionally, for any given lamp type, there will be proprietary differences from manufacturer to manufacturer in arc tube geometry, arc chemistry, envelope dimensions, and internal frame and thermal baffle construction that can influence light output, correlated color temperature, and the spatial distribution of energy from the lamp.

Through standards activity such as from ANSI, manufacturers arrive at agreements on electrical and opto-mechanical features such as starting voltages and light center lengths of lamps to create socket-level compatibility from manufacturer to manufacturer. Regardless, the possibility exists that the luminaire will be operated and measured with a lamp different than what was used in the photometry and the application design. It is important to verify that the lamps used for field measurement are accounted for appropriately in the performance predictions. When the installed

lamp is determined to be different from that used for performance predictions, care should be taken to consider the potential influences on light output and measured performance.

2.2.1.1 Manufacturing Variations. The light output of lamps in terms of color, initial lumen output, and rates of lumen depreciation, even within a single production lot from a single manufacturer, will vary. The variation in light output is a natural result of variation in the components and processes used to manufacture HID lamps. Some examples of lamp manufacturing variables include halide doses, electrode spacing, arc tube diameters, pinch location, end-coat coverage, electrode metallurgy, getter composition, and jacket bake-out time. Manufacturers have developed component tolerances and process capabilities to reduce the variation in the light output of HID lamps. However, as a practical or economic matter, some variation will remain.⁵ Additionally, electrically interchangeable HID lamps from different manufacturers can have different light output due to design and process differences from manufacturer to manufacturer.

A manufacturer's ratings for initial lumens and lumen depreciation are typically established by statistical methods to ensure that the majority of lamps produced meet rated light output. During development and initial production of a lamp, samples are run for thousands of hours and measured at regular intervals to develop statistics that describe the lamp's light output performance. However, since it is a statistical method, it is also likely that the performance of some fraction of the lamps produced can vary significantly from the rated initial output or the mean lumen maintenance curve for that type of lamp. What fraction differs significantly from specification depends on the manufacturer's choices for establishing ratings versus the test statistics.

Finally, HID lamps are typically produced in compliance with ANSI standards⁶⁻⁸ on electrical parameters such as arc voltage, and geometric properties such as light center, eccentricity, and overall dimensions.^{6,7} These standards are used to establish conditions for product interchangeability between manufacturers. Inherent in the standards are tolerances on all the controlled parameters. Like the light output properties, the

electrical and geometric properties can be influenced by the statistical nature of manufacturing process capability as it relates to these tolerances.

2.2.1.2 Light Output at Field Installations. Several key factors can influence the measured light output in field installations. The initial lumen rating of HID lamps is based on 100 hours of operation. Lumen output can exceed the catalog rating by 10% to 30% for periods of less than 100 hours. The lumen output will depreciate during the burn-in time and stabilize near the rated initial lumen value. Light output also decreases with operating hours on the lamp. A plot of lamp lumens versus operating hours is called a *lumen maintenance curve* and is available from lamp manufacturers. The operating hours per lamp start and the installed tilt also influence the expected light output during the lamp's operating life. Manufacturers may provide ratings and adjustment factors to lumen depreciation rates for tilted operation and for hours per start that differ from the lamp rating conditions.

For more accurate field measurements, the lamp manufacturer's recommended burn-in time will need to be followed. When taking initial lighting measurements, allowance should be made for complete flux stabilization after the installation of the system. The lumen output of many HID lamps is orientation (i.e., tilt) sensitive. An orientation (tilt) change, even after a 100-hour burn-in, will cause some lumen output change until the halides redistribute and re-stabilize in the arc tube. For major orientation (tilt) changes (greater than approximately 30 degrees), a re-stabilization period of as much as 48 hours may be required to redistribute the halide pool and stabilize the lamp lumen output. For minor orientation (tilt) changes (shifts less than approximately 30 degrees), a stabilization period of 4 to 6 operating hours is recommended before measurement. When measuring systems with significant operating hours, knowledge of the actual hours of lamp operation, typical hours per start, and average installed tilt of the lamps may be necessary to correlate predictions with actual field performance.

2.2.2 Arc Tube Alignment. All lamps have component and assembly tolerances that allow some variation. The allowed tolerances for most lamps are controlled by ANSI

standards agreed to by multiple lamp manufacturers. Lamps selected as "standard" samples for photometric laboratory tests are typically chosen to be as close to the "nominal" ANSI geometric specifications as is practical. In addition, care is normally taken to center the lamp in the optical system for photometric measurement. Typical production lamps installed in the field could have variations in the position and dimensions of the arc with respect to the lamp base and optical system.

The effect of these variations in the field depends on the characteristics of the luminaire, the luminaire optical system, and the lamp. In some cases, more light may be delivered in one quadrant of the output for symmetrical or asymmetrical types of luminaires as compared to the other quadrants. Such variations might cause the measured uniformity to deviate from prediction. In some instances, the position shifts might cause fixture efficiency changes, raising or lowering the expected light levels. With HID lamps that have known sensitivity to reentrant radiation, such as HPS, the position shifts might change radiant loading on the arc tube, modify the arc voltage rise, and change the lamp's lumen output. Any of these effects can contribute to the field results having some variation from the predicted values.

2.2.2.1 Light Center and Arc Length Variations. In HID lamps, arc length and the distance between electrodes can vary. Per ANSI standards, the light center length (LCL; the distance from geometric center of the arc tube to the base) has location tolerances.^{6,7} The combination of allowed tolerances per ANSI standard on linear dimensions alone can result in the arc center location deviating from nominal position by up to 6 mm and the nominal arc length by up to 7 mm for typical 400-watt ED37 or BT37 HID lamps (see **Figure 2-1**).⁸ Increases in the arc length could, for example, depending on the lamp orientation, broaden the angular extent of vertical or horizontal lobes in the photometric pattern, and change the maximum candela value in comparison to the laboratory test conditions. Similarly, variations in the light center length (LCL) would shift the vertical or horizontal angular positions of photometric pattern features by changing the arc position relative to the reflector or refractor components of the luminaire. In both cases, the associated changes in the photometric patterns could result in light level and uniformity

differences between the predicted values based on laboratory measurements using lamps selected for “nominal” light center and arc length dimensions, and the values obtained from field measurements using “typical” lamps.

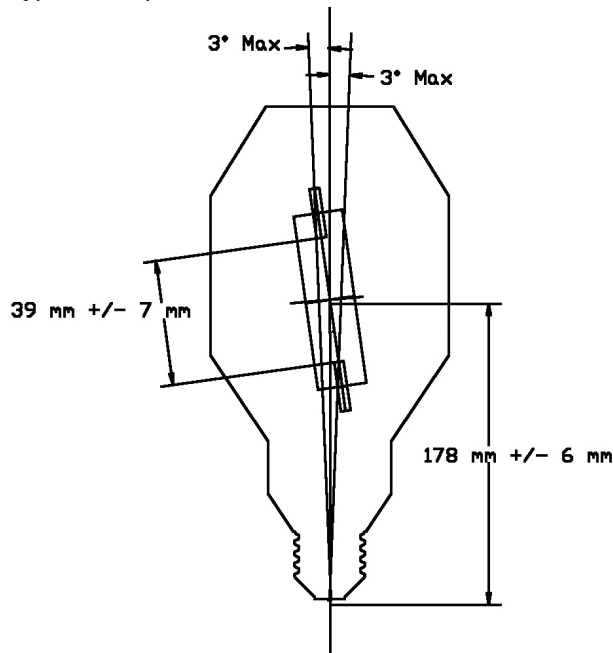


Figure 2-1. Linear and angular specifications from ANSI C78.1375-1996 for a 400-watt ED37 or BT37 MH lamp.

2.2.2.2 Axial Position of Arc Tubes and Lamp Bases.

The axial position of the arc tube is also controlled by ANSI standard through angular tolerances on electrode tip variation to the lamp base axis. The typical tolerance is 3 degrees of electrode tip variation with respect to the axis of the lamp base.⁸ This angular tolerance, when combined with the nominal light center length (LCL) tolerances and with similar tolerances for the eccentricity of the lamp base, can result in arc center location deviating from nominal position by over 8 mm for typical 400W ED37 or BT37 jacketed HID lamps. Variation in the axial position of the arc tube and associated light center can change the light center position in comparison to the laboratory test conditions using lamps selected for nominal arc tube geometry. Changes in light center position relative to the reflector or refractor components can change the angular positions of photometric pattern features and, in some instances, the efficiency of the luminaire. Either of these changes could result in field measurements of light levels or uniformity that deviate from predicted values.

2.2.3 Arc Tube Temperature. HID lamps can exhibit a difference in operation in free air compared to a hotter condition inside an enclosed optical system. In the case of HPS and LPS lamps, the operating arc tube voltage tends to rise when either of these conditions occurs:

- The temperature of the lamp is raised because of the enclosing optical system.
- The lamp’s radiant energy is redirected by the reflector through the arc tube.

Each of these factors results in a higher arc tube voltage, which further increases as the lamp ages, until the point is reached where the ballast no longer provides sufficient voltage to maintain the arc. At this point, the HPS lamp is said to “drop out.” As the arc tube cools, the voltage available from the ballast is adequate to restart the lamp. As it warms up again, the lamp will again drop out. This “cycling” generally signals the normal end of life for HPS lamps.

2.2.4 Position Factor. Some lamps are designed to burn in any position (i.e., “U” or “Universal” lamps), while other lamps are designed to burn in specific positions. To obtain rated life and rated light output, lamps designed for VBU (vertical base up), VBD (vertical base down), or HOR (horizontal) operation should be burned in the stated position.⁹ MH lamps designated for operation over a range of burning positions will have a different light output (on the same ballast) for virtually every change in lamp angle (VBU, VBD, or HOR).

Lamp position factors, sometimes termed “tilt factors,” are available from lamp manufacturers’ data. This data is usually provided as a plot or tabulation of the lamp’s luminous output as a function of orientation from “vertical base up” (VBU), typically designated as 0 degrees, to a lamp orientation of “horizontal” normally designated as 90 degrees. These factors can be averages for several MH lamp types and care should be taken in interpreting and applying the factors to ensure that they are the best representation of use. **Figure 2-2** illustrates a conceptual tilt factor curve for a MH lamp. Individual lamps within a specific type may vary significantly from average values. Lumen output variations for Hg lamps with different operating positions also occur, but they are more consistent and repeatable than the MH output variations. HPS and LPS

lamps also exhibit position-sensitive operation, but MH lamps evidence the greatest light output variation when tilted from rated position.

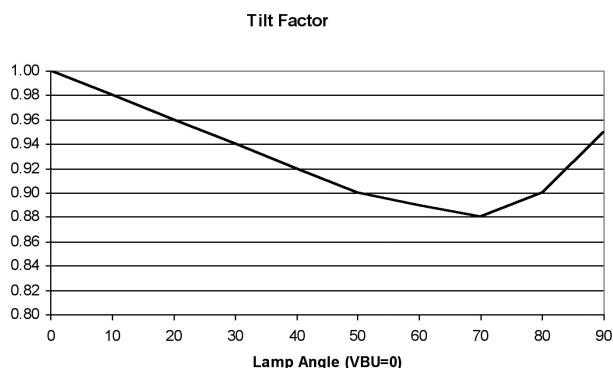


Figure 2-2. Example tilt factor curve for a MH lamp.
(Note: Graph is for illustrative purposes and should not be considered representative of actual lamp performance.)

2.2.5 Special Purpose Lamps. Lamp manufacturers produce a large variety of HID “special purpose” lamps, including lamps with optical elements such as PAR lamps; base or jacket configurations (e.g., bi-pin, double-ended, and position oriented); spectral variants such as color-dosed lamps; and “open-rated” or “protected” versions of standard lamps. These lamps can present issues with respect to prediction versus measurement in the field that differ from those of “general purpose” HID lamps of conventional construction and basing. Examples include:

- Color-dosed lamps can create issues with respect to the photopic response-matching needs of the light meters used. As the spectral width of the light source narrows, the contribution to the light meter’s f_1' values, over the wavelength range of interest are required to be sufficiently low to ensure acceptable uncertainty in the measured light levels.¹⁰
- Lamps, such as PAR types, that incorporate prismatic or lensed refractors and integral reflecting surfaces for the purpose of beam forming, contain sources of variation that parallel “luminaire” factors for the same types of components. In general, the construction and fabrication methods for lamp-integral refractors and reflectors should exhibit less variability than the luminaire equivalents, but they do exist as factors that could influence measured results. Since PAR lamps are essentially integrated floodlights, the tolerances on beam width and beam aiming could be too variable to ensure tight tolerances on light levels in an application.

- “Special” lamp bases and jacket configurations create differences in the tolerance stack-up associated with location of the light center in a luminaire. These bases are also industry standards, but differ from conventional medium or mogul based lamps with BD, ED or BT jacket types. In some instances, the light center accuracy may be better. Regardless, it is important to recognize how the socket and base configuration relates to output variability in the luminaire photometry.
- Unjacketed tubular double-ended lamps, as may be used in some sports lighting luminaires, can operate at significant efficacy differences from published data if the thermal environment in the luminaire does not adequately mimic the conditions created in the laboratory for the lamp’s lumen rating.
- “Position-oriented” and “protected” (i.e., exclusionary base) lamps can sometimes be inserted in conventional sockets. Protected lamps with exclusionary bases will fit in standard sockets but have output lumen ratings that are typically lower than their non-protected equivalents. Conventional, non-oriented lamps can be inserted in position-oriented sockets (e.g., POMB sockets), resulting in the possibility that an unplanned lamp will be used in a luminaire.

When troubleshooting an installation or trying to reconcile measured versus predicted values for installations using “special purpose” lamps, it is important to recognize that special factors may be present and to review how they might influence the lighting results.

2.3 Luminaire Optical Variations

2.3.1 Reflector Materials. Luminaire photometric properties depend on the combined specular and diffuse reflectivity properties of the materials used in the luminaire’s optical system design. A wide variety of materials such as aluminum sheet, paint, injection molded plastics, and vapor deposited coatings serve the function of luminaire reflectors. The materials and values listed in this section are provided as illustrations and do not represent the specific properties of any given material.

Aluminum reflector materials are available that range from specular to diffuse, with total reflectance levels as high as 95%. In addition to simple sheet stock, aluminum reflector materials are available with chemically and mechanically produced textures to create various ratios of diffuse to specular reflection and controlled scattering properties. Painted reflector finishes are available with total reflectance up to 96%, with surface properties ranging from high gloss to matte.

For many materials, some change in specular or diffuse reflectance can occur from batch to batch or over time. Variations in the reflector material's optical characteristics that significantly change the specular or diffuse reflectivity levels or the spatial distribution of its scattering properties can result in corresponding variations in luminaire light output and photometric shape. Variation in the reflector material's production processes could result in production luminaires with reflectors having small but potentially significant differences in the total reflectivity or scattering when compared to the sample used for photometric testing in the laboratory.

2.3.2 Reflector Shape. Some preliminary photometric data might exist on prototype reflectors produced on temporary tooling or fabricated from prototype parts; as such, these should not be used in preparing a lighting design. Permanent tooling used for production runs could alter the actual total efficiency or change the distribution of the light as compared to the prototype model.

Many reflectors consist of "spinnings." If forming tolerances are not carefully maintained, variations in surface quality and "spring back" could alter the candela distribution and total luminous output of the luminaire. Manufacturers should periodically perform photometric measurements on representative samples of production parts when the photometric data is for publication.

2.3.3 Refractors and Enclosures. The optical characteristics of glass and plastic refractors and enclosures can show variation for a variety of reasons. Part shapes and small-scale features will change as the molds wear with use. The refractive index of glass and plastic can change slightly from lot to lot, and the glass or plastic can exhibit differences in the amount of voids,

bubbles, and inclusions from batch to batch and part to part. These imperfections can result in changes in the light distribution from the original design. Some plastic refractors can exhibit changes in material tolerances, reduced mechanical strength, and reduced transmission caused by degradation from operating temperature and ultraviolet exposure over time.

2.3.4 Luminaire Assembly Tolerances. Luminaire components are assembled in a variety of ways. The typical luminaire assembly incorporates a reflector, glass or plastic enclosure or refractor, lamp, socket, and lamp socket bracket, all positioned relative to each other by direct connection or through the luminaire's housing components. Each component in this chain has production tolerances and allowed variation on dimensions and mounting features. Assembly variations that result in relative position differences in the optical components can cause differences in the photometric performance of the luminaire.

2.3.5 Luminaire Configurations and Options. Many luminaires are sold with interchangeable and field configurable options that are significant with respect to the photometric properties of the luminaire. Examples of such options include field replaceable reflectors, internal or external house-side shielding components, field rotatable reflectors to allow flexibility in optical and pole orientations, polycarbonate vandal shields, and lamp positions. It is important to confirm that the installed configuration matches the configuration used for application simulation and performance predictions. Many of these options can be verified via ground level visual inspection of the mounted luminaire compared to the design plan and the manufacturer's catalog. However, some options such as lamp position may require internal inspection of the luminaire's optical components to verify the installed versus designed condition.

2.4 Field Conditions

2.4.1 Mounting Height. Illuminance at a point on a surface normal to the direction of incident light varies directly with the luminous intensity of the source and inversely as the square of the distance between the source and the point of measurement. The use of

pole heights instead of actual luminaire mounting height can introduce errors. Actual mounting heights of the luminous opening should be measured when the illuminance or luminance is recorded so as to report the actual field conditions.¹¹

2.4.2 Spacing. Deviations from the design spacing will affect the uniformity of the lighting job as well as the average illuminance.¹¹ If poles in outdoor installations or suspension points from ceilings in indoor installations are closer than the designed spacing, average illuminance will be higher than predicted. Increases in the spacing will reduce the average illuminance.

2.4.3 Aiming and Tilt. Field conditions and the actual position of the luminaires might not be as exact as in the application design; therefore, location and positioning of units where field measurements and tests are being made should be verified.¹¹ In floodlighting design, aiming may be accomplished by selecting a vertical and a horizontal degree setting for each luminaire. If cross-arms are not exactly in their intended position and the poles not precisely located and in a vertical position, errors can occur that will cause luminance and illuminance variations. The angles shown on mass-produced vernier scales inscribed on luminaires could exhibit an inaccuracy of from one to five vertical degrees. Inaccuracy of these devices might cause erratic aiming by the installers. In addition, pole position and luminaire aiming can change after years of use. As noted in the discussion of lamp factors (see **Section 2.2 Lamp Variations**), the expected light output of HID lamps varies with lamp orientation changes, and this could be caused by unexpected luminaire tilts.

2.4.4 Topography and Obstructions. Where the ground or roadway surface to be lighted has significant contours, such as hills or roadbed banking, it is difficult to identify the reference surface used for design calculations. Similar situations might be found on golf courses or ski slopes. Lighting design software programs typically provide a flat metering plane and the ability to tilt the plane but do not always offer computation on a fully contoured surface. These “simplifications” can result in design assumptions differing from actual mounting heights by several feet in cases where significant contours exist. Some types of installations

might require that a continuously variable reference surface be used in the design calculation.

In many instances, obstructions need to be considered as a component of the lighting design. On occasion, the presence of obstructions can be overlooked in the design and predictive analysis, due to incomplete, inaccurate, or out-of-date site plans. Care should be taken to review the design plan as compared to the actual site conditions before measurement, to verify that any obstructions on the site have been accounted for in the predicted lighting performance. In cases where obstructions that could affect measured values have been overlooked, a revised predictive model for the site is recommended.

2.4.5 Surface Properties. Illuminated surfaces in proximity to the area of measurement can significantly affect the lighting performance of an installed system. Lumens reflected from or absorbed at these surfaces can affect light levels and uniformity and should be considered in field measurements and predictive analysis. In cases where the predictive performance analysis has included the effects, the potential exists that the surface properties in the field differ from the assumptions of the design (e.g., the design assumed a natural brick surface, but someone decided to paint it white). In cases where the surfaces have not been considered in the predictive work, efforts to isolate and quantify the contributions from these surfaces in the field or access to a revised predictive model are necessary to account for the effects.

2.5 Field Measurements

2.5.1 Meter Accuracy. A clear and concise listing of all pertinent factors and conditions during field measurements is essential. Items listed throughout this document, such as recording the voltage, temperature, and condition of the system are necessary to identify causes of abnormal measurements. The meter type and model should be recorded as well as the date of calibration. Periodic meter calibration is important to the accuracy of reported data. In general, it is recommended that meters be calibrated on an annual basis unless their calibration stability is demonstrated to be longer than a year.

Multi-meters used for field measurement of electrical parameters should have an accuracy of 0.5% or better. Electrical meters should respond to the true root-mean-square (RMS) values, as luminous output is a function of power.

Light meters are required to use a detector-filter combination that matches the CIE photopic response curve with an f_1' value of 5% or better and be “cosine-corrected” with an f_2' value of 5%.¹² Quality light meters used in the laboratory and many portable hand-held meters are frequently certified by their manufacturers to “3.0 percent accuracy” or better. However, care should be taken in the interpretation of the published accuracy information to ensure that the published accuracy measurements meet the intent of the CIE’s f_1' and f_2' measures, to ensure the proper spectral sensitivity weighting and angular sensitivity of the meter. Even with meters having the above mentioned characteristics, the total measurement uncertainty should be considered when evaluating the results of a field measurement with respect to predicted levels.

As with any light measurement, it is critical that care be taken with respect to extraneous light sources and the shadowing effects of obstructions. When taking light meter readings in the field, the operator should stand so that shadows or reflections do not fall upon the meter. The choice of operator clothing is preferably dark to avoid reflected light that could affect the measurements. Finally, when a lighting installation is measured, daylight will need to be excluded unless it is part of the design.

2.5.2 Contaminant Conditions. Lighting performance varies as a system ages and is exposed to contaminants. Contaminants may include airborne particulates (e.g., dirt), gases, and liquids of various types. If field measurements are made a short time after the luminaires are put into service, contaminants may not have accumulated on external or internal parts of the luminaires. However, over time, deposits are possible as even a “sealed” luminaire will “breathe” as a natural function when turned on and off. Contaminants impact the lighting performance in more complex ways than simple transmission loss. Most contaminants also act as scattering media on optical surfaces; thus, they can

change the photometric distribution and, especially, the peak candela values. Prior to field measurements, the luminaires should be inspected so that their conditions can be noted as part of the recorded data.

2.5.3 Light Meter Positioning. The light meter should be oriented with its light-sensing surface consistent with the measurement being taken. For measurements of horizontal light levels, the light meter should be held horizontal and at a height consistent with the intended lighting application. For measurements of vertical light levels, the light meter should be held in a vertical plane at the height specified in the design specifications. In cases of measuring “vertical” levels for television broadcasting, it may be necessary to “aim” the light meter toward the intended camera positions.

2.5.4 Measurement Grid and Registration. Lighting designs are typically created and evaluated using computer based applications software. The software uses user-defined or predefined measurement grids as simulated metering points to compute the maximum, minimum, and average light levels, and various uniformity parameters. Field measurements are made by physically placing a light meter at the design’s intended measurement grid points and measuring the light levels for each point at the site.

Useful comparisons of measurement to prediction depend on both the design and the measurement data representing the same points on the site. Errors in the assumed height of the measurement grid, the x and y grid intervals, or the registration of the grid to the site can cause differences between measurement and prediction. A site’s sensitivity to meter positioning error is proportional to the error’s influence on the assumed distance from the sources that contribute to the light level at that metering point. For example, if a measurement is being made to confirm a nadir light level under a single luminaire mounted at 10-meter height with a planned grid height of 1 m above ground level, and the measurement data is accidentally taken at 0.8 m above ground level, the error at nadir would be: $1 - [(10 - 1)^2 / (10 - 0.8)^2] = 4.3\%$.

For a metering point with multiple contributors, the measurement error is the average of the errors for each

source. While this is a simplistic example, it illustrates that the errors can be significant, and it is important that care be taken to limit the possibility of grid and grid-registration errors. Grid and grid-registration accuracy requirements are site specific and merit evaluation before the measurements are conducted.

2.5.5 Humidity and Temperature. Light meters can be sensitive to both humidity and temperature. Meter manufacturers typically specify an acceptable operating temperature and humidity range over which the instrument's accuracy is guaranteed. For example, a meter might be specified for an operating temperature range of "-10 °C to +35 °C" and a humidity limit of "<80% RH Non-condensing." The high humidity conditions associated with rainfall could exceed the manufacturer's specifications for the instrument and affect the high-impedance electronic circuitry of a light meter. Older barrier-layer selenium photodetectors can also exhibit sensitivity to high humidity conditions that would affect responsivity and calibration. Some light meters, particularly older types, can be sensitive to temperature variations due to temperature coefficients in electrical components such as feedback resistors, or temperature coefficients of offset current and voltage in the amplifier circuitry. Field measurement conditions should be reviewed with respect to the specified operating limits of the instrumentation to ensure that the field conditions during measurement are within the manufacturer's limits.

2.5.6 Atmospheric Transmissivity. It is important that attention be given to atmospheric transmissivity, particularly when evaluating installed systems such as searchlight installations and ascertaining that additional measurements are required. Fog and haze conditions that may just be barely perceptible visually can have significant effects on photometric measurements.

2.5.7 Weather and Site Conditions. A variety of weather and site-specific conditions can contribute to differences between measured and predicted light levels. Sources of extraneous light, such as direct or reflected spill light from adjacent lighting installations, can artificially raise measured light levels and affect the measured uniformity. Similarly, background light levels due to cloud-reflected light from the surrounding

community or moonlight contributions can influence measured light levels. When possible, measurements should be made with the stray light contributors turned off, or baseline measurements of the background light level contributors should be made to account for their effects on measured results. Local and temporary conditions such as light fog can have a tremendous negative impact on measured light levels and measured uniformity due to the atmospheric scattering created by these conditions. Care should be taken to avoid these weather and site conditions during measurement whenever possible and to note and account for these factors when avoiding them is not possible.

3.0 General Considerations

Many factors affecting the correlation of laboratory measured photometry and design prediction with the reality of luminaire installations and field measurements have been discussed. There remain other factors that deserve general consideration.

Uncertainty exists to some level in any measurement. The laboratory practices detailed in various IES Lighting Measurement (LM) documents attempt to place an upper bound on this uncertainty by establishing good practice and setting accuracy standards for instruments used in the measurements. As a general rule, the photometry as performed in different laboratories under IES recommended conditions are intended to agree within 2%. In reality, laboratory-to-laboratory correlation as good as 2% can be difficult to achieve, given the realities of luminaire alignment, test lamp geometry, lamp positioning, and electrical conditions. Efficiency differences on the order of 1% to 3%, and candela value differences of 5% to 10% for any given polar angle, could be expected in laboratory-to-laboratory comparisons. Field measurements are unlikely to match computer predictions based on laboratory data better than the expected laboratory-to-laboratory correlation for a single luminaire.

With the number of energy conserving lamps and ballasts that are interchangeable with each other or

with standard components, an element of uncertainty has been introduced into the design process. The photometric test results for a luminaire are no longer independent of lamp and ballast type, because of changes in luminaire heating and the interaction of some lamp-ballast combinations. Ballast factors might increase or decrease. In lieu of better methods, the only way to reduce uncertainty in the design process is to submit the luminaire with its particular lamp-ballast combination for photometric testing.

Many of the issues discussed in this LM are relevant to or have parallel issues for indoor HID luminaires and indoor HID applications. This LM does not specifically discuss the indoor application factors, but it should be recognized that many of the factors exist and merit consideration.

Measurement formats and reporting recommendations are extensively treated in other IES LMs and are not duplicated in this LM. **Annex A** provides general information relevant to the recording and reporting of field data.

4.0 Additional Reading

Lewin I, Stafford L. Real world use of photometric test reports. *Light Des Applic*. 1988 Jan;18-23.

Lewin I, Stafford L. Photometric and field performance of metal halide luminaires. *J Illum Eng Soc*. 1988 Winter;15-19.

Lewin I, Stafford L. Photometric and field performance of high pressure sodium luminaires. *J Illum Eng Soc*. 1987 Summer;106-14.

Annex A – Measurement and Recording Recommendations

Field measurement and recording recommendations vary by application, but in many cases there is basic

information that should be collected as standard practice regardless of the lighting application type. Test reports should include:

- Location, with adequate detail to ensure proper identification of the site
- Date and time of the measurements
- A detailed description of the installation and lighting equipment, including pole spacing, mounting heights, aiming angles and other relevant conditions
- Names of all witnesses taking measurements
- Age of the lighting installation
- Description of the measurement instruments used and calibration date
- Weather and site conditions
- Electrical operating conditions, including supply voltages and an indication of where they were measured relative to the lighting system electrical distribution
- A summary of any special circumstances, such as unplanned obstructions, surrounding conditions, or extraneous light sources
- A description of the luminaire condition, such as contaminants and state of repair
- Documentation of the measurement grids and registration to the site plan
- Height and orientation of the measuring instruments
- A tabulation of the measured data
- Maximum, minimum, and average light levels per area or planned sub-area in the site
- Uniformity values such as maximum-to-minimum (max:min) and average-to-minimum (avg:min) ratios and coefficient of variance per area or planned subarea in the site

Annex B – Measured vs. Predicted Performance Checklist

	Looked At	Needs Review		Looked At	Needs Review
Power Supply and Ballast Variations			Position Factor	☐	☐
Power Supply Voltage	☐	☐	Special Purpose Lamps	☐	☐
Distribution System			Luminaire Optical Variations		
Impedance	☐	☐	Reflector Materials	☐	☐
Wave Shape of Voltage and Current	☐	☐	Reflector Shape	☐	☐
Ballasts	☐	☐	Refractors and Enclosures	☐	☐
Design Variations	☐	☐	Luminaire Assembly		
Core and Coil	☐	☐	Tolerances	☐	☐
Capacitor	☐	☐	Luminaire Configurations and Options	☐	☐
Starting Aids	☐	☐	Field Conditions		
Wave Shape of Voltage and/or Current	☐	☐	Mounting Height	☐	☐
Ballast Component			Spacing	☐	☐
Temperatures	☐	☐	Aiming and Tilt	☐	☐
Lamp Variations			Topography and Obstructions	☐	☐
Light Output of the Lamp	☐	☐	Surface Properties	☐	☐
Manufacturing Variations	☐	☐	Field Measurements		
Light Output at Field			Meter Accuracy	☐	☐
Installations	☐	☐	Contaminant Conditions	☐	☐
Arc Tube Alignment	☐	☐	Light Meter Positioning	☐	☐
Light Center and Arc Length Variations	☐	☐	Measurement Grid and Registration	☐	☐
Axial Position of Arc			Humidity and Temperature	☐	☐
Tubes and Lamp Bases	☐	☐	Atmospheric Transmissivity	☐	☐
Arc Tube Temperature	☐	☐	Weather and Site Conditions	☐	☐

REFERENCES

- 1 Illuminating Engineering Society. ANSI/IES LS-1-20, Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: <https://www.ies.org/definitions/>. (Accessed 2019 Sep 13).
- 2 Unglert MC, The need for high pressure sodium ballast classification. Light Des Appl. 1982;13(3):29.
- 3 National Electrical Manufacturers Association (NEMA). ANSI C82.5-2016. American National Standard for Reference Ballasts – High-Intensity-Discharge and Low-Pressure Sodium Lamps. New York: ANSI; 2016.
- 4 Ward PC. Estimation of HPS lamp voltage rise as a function of ballast characteristic curve parameters. J Illum Eng Soc. 1983;13(1):157.
- 5 Levin RE. The Photometric Connection, Parts 1 through 4. Light Des Appl. 1982 Sep; 1982 Oct; 1982 Nov; 1982 Dec.
- 6 National Electrical Manufacturers Association (NEMA). ANSI C78.386-1989 (R1994), American National Standard for Electric Lamps – Mercury Lamps – Methods of Measuring Characteristics. New York: ANSI; 1994.
- 7 National Electrical Manufacturers Association (NEMA). ANSI C78.389-2004, American National Standard for Electric Lamps – High Intensity Discharge – Methods of Measuring Characteristics. New York: New York: ANSI; 2004.
- 8 National Electrical Manufacturers Association (NEMA). ANSI C78.1375-1996, American National Standard for Electric Lamps 400-Watt, M59 Single-Ended Metal-Halide Lamps. New York: ANSI; 1996.
- 9 National Electrical Manufacturers Association (NEMA). ANSI C78.380-2016, American National Standard for Electric Lamps – High-Intensity Discharge Lamps – Method of Designation, New York: ANSI; 2016.
- 10 International Commission on Illumination (CIE). 1987. CIE 69:1987, Methods of Characterizing Illuminance Meters and Luminance Meters: Performance, Characteristics and Specifications. Vienna: CIE; 1987.
- 11 Odle HA. HID field measurements vs. data calculated from photometry. J Illum Eng Soc. 1975;4(2):84.
- 12 Illuminating Engineering Society. ANSI/IES LM-28-20, Approved Method: Guide for the Selection, Care, and Use of Electrical Instruments in the Photometric Laboratory. New York: IES; 2020.



Lighting Science Standards

Fundamentals, Metrics and Calculations

Lighting Practice Standards

Design, Engineering, and Specifications

Lighting Applications Standards

Design Criteria and Illumination Recommendations

Lighting Measurements and Testing Procedure Standards

Industry Standardization

Roadway and Parking Facility Lighting Standards

Criteria and Illumination Recommendations

Order# ANSI/IES LM-61-20(R2023)

ISBN# 978-0-87995-212-9

www.ies.org